Basis and dimension.

Video 12.1.1

Consider a subspace $V \subset \mathbb{R}^n$.

A *basis* for V is a list of vectors in V which is a spanning set for V and which is "as short as possible".

Definition ('Basis for a subspace').

Consider some Euclidean space $\mathbb{R}^n$ and some subspace $V \subset \mathbb{R}^n$.

Consider some finite list of vectors in this V:

$$\vec{v}_1, \dots, \vec{v}_m \in V.$$

This list of vectors is said to be a **basis** for V if

1. $V = span\{\vec{v}_1, \dots, \vec{v}_m\}$
2. The list $\vec{v}_1, \dots, \vec{v}_m$ is linearly independent.

This says: Every vector in V can be built as a linear combination of this set of "building blocks" …

… and this is the shortest possible list of vectors with this property.

Discussion:

For a list $\vec{v}_1, \dots, \vec{v}_m \subset V$ to be a basis of V, it must satisfy two requirements:

Requirement #1: The list must span V.
Requirement #2: The list must be linearly independent.

Note that these two requirements *compete* with each other.

Spanning set for V. ⟵ Possible basis $\vec{v}_1, \dots, \vec{v}_m$. ⟶ Linearly independent set.

To satisfy the spanning set condition you would make the set larger…

… whereas to satisfy the linearly independent condition you would make the set smaller.

A basis ***balances*** these two requirements.

The following example is the most important one.
This is the example this general definition of a basis generalizes.

Example (The standard basis of $\mathbb{R}^n$.)

Fix some $\mathbb{R}^n$.
Consider the subspace consisting of the full subset $\mathbb{R}^n \subset \mathbb{R}^n$.
The **standard basis** of $\mathbb{R}^n$ is the following list of vectors

$$\vec{e}_1, \dots, \vec{e}_n$$

where if $1 \le i \le n$ then $\vec{e}_i$ is the vector which is zero everywhere except for a single 1 in the i-th position

$$\vec{e}_i = (0, \dots, 0, 1, 0, \dots, 0)$$

Position i.

For example: in the case $\mathbb{R}^4$ the basis is

$$\vec{e}_1 = (1,0,0,0), \vec{e}_2 = (0,1,0,0), \vec{e}_3 = (0,0,1,0), \vec{e}_4 = (0,0,0,1).$$

show
$B = \{(1,0,0,0), (0,1,0,0), (0,0,1,0), (0,0,0,1)\}$
forms a basis in $\mathbb{R}^4$

① $\mathbb{R}^4 = \text{span}(B)$

Let $\vec{v}$ be an arbitrary vector in $\mathbb{R}^4$.
$\vec{v} = (v_1, v_2, v_3, v_4)$
$= v_1(1,0,0,0) + v_2(0,1,0,0) + v_3(0,0,1,0) + v_4(0,0,0,1)$

② To show B is linearly independent, must show the only c_i's solving
$c_1(1,0,0,0) + c_2(0,1,0,0) + c_3(0,0,1,0) + c_4(0,0,0,1) = (0,0,0,0)$ are $c_1 = c_2 = c_3 = c_4 = 0$ → trivial sol^{n}

LHS $= (c_1,0,0,0) + (0,c_2,0,0) + (0,0,c_3,0) + (0,0,0,c_4)$
$= (c_1, c_2, c_3, c_4)$

$\therefore \begin{cases} c_1 = 0 & c_3 = 0 \\ c_2 = 0 & c_4 = 0 \end{cases}$

Why does this form a basis?

Following the definition of basis, there are two things we have to explain.

First part of the definition:
Why does the set $\{\vec{e}_1, \dots, \vec{e}_n\}$ span $\mathbb{R}^n$?

Answer:
We must explain why every vector in $\mathbb{R}^n$ is equal to some linear combination of the vectors $\{\vec{e}_1, \dots, \vec{e}_n\}$.

Let $\vec{v}$ be an arbitrary vector in $\mathbb{R}^n$. There will be some scalars $v_1, \dots, v_n$ such that $\vec{v} = (v_1, \dots, v_n)$.

We can split up this vector component-by-component, to get:
$$\vec{v} = v_1(1,0, \dots, 0,0) + \cdots + v_n(0,0, \dots, 0,1).$$

This equation directly shows what we want: that $\vec{v}$ is a linear combination of $\vec{e}_1, \dots, \vec{e}_n$.

Why does this form a basis?

Following the definition of basis, there are two things we have to explain.

Second part of the definition:
Why is the list $\vec{e}_1, \dots, \vec{e}_n$ linearly independent?

Answer:
According to the definition of linear independence, we need to show that the only list of scalars c_i solving the equation
$$c_1\vec{e}_1 + \cdots + c_n\vec{e}_n = \vec{0} \quad (\star)$$
is the trivial solution $c_1 = c_2 = \cdots = c_n = 0$.

So we need to find the set of solutions to equation $(\star)$.

First remember what each $\vec{e}_i$ refers to. The equation becomes:
$$c_1(1,0, \dots, 0) + \cdots + c_n(0,0, \dots, 1) = (0,0, \dots, 0).$$

Second part of the definition:

Why is the list $\vec{e}_1, \dots, \vec{e}_n$ linearly independent?

Answer (continued):

Computing the LHS we get:

$$(c_1, c_2, \dots, c_n) = (0,0,\dots,0).$$

This is a system of linear equations. We get one equation for each position by equating corresponding entries on the two sides. The system is:

$$\begin{cases} c_1 & = & 0 \\ & \vdots & \\ c_n & = & 0. \end{cases}$$

Thus there is only one solution, the trivial solution:

$$c_1 = c_2 = \cdots = c_n = 0.$$

Thus the list $\vec{e}_1, \dots, \vec{e}_n$ is linearly independent.

$\square$

<u>*Example*</u>

Video 12.1.2

Show that the following list of vectors gives an alternative basis of $\mathbb{R}^3 \subset \mathbb{R}^3$:

$$(1,1,1), (1,2,3), (1,0,0).$$

<u>*Solution*</u>

There are two parts to the definition of basis.

We need to check:

1. These vectors span $\mathbb{R}^3$.
2. These vectors are linearly independent.

We have learnt some matrix algorithms for answering these questions. We'll check them in turn.

Show that the following list is a basis for $\mathbb{R}^3$.
(1,1,1), (1,2,3), (1,0,0)

① Must show this list spans all of $\mathbb{R}^3$

② And must show it is linearly independent

① $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 3 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & -1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Because the row echelon form has no row of zeros, the given list of vectors spans all of $\mathbb{R}^3$.

② $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Because this row echelon matrix has a leading entry in every column, the given list is linearly independent.

Thus it is a basis.

Solution

Question 1: Do these vectors span $\mathbb{R}^3$?

Recall the algorithm we learnt to check this during the "spanning vectors" topic:
1. Construct a matrix whose columns are the given vectors.
2. Do Gaussian elimination to turn this into row echelon form.
3. If there are no all-zero rows then yes the vectors span $\mathbb{R}^3$.

Let's do it:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 3 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

This row echelon form has no all zero rows so the three given vectors do indeed span all of $\mathbb{R}^3$.

Solution (continued)

Question 2: Are these vectors linearly independent ?

We also learnt an algorithm to check this.
1. Form a matrix whose columns are the given vectors.
2. Do Gaussian elimination to turn it into row echelon form.
3. If every column has a leading entry then the vectors are linearly independent.

We already have the row echelon form: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$.

Every column has a leading entry so the given vectors are linearly independent.

In summary: the given vectors (i) span all of $\mathbb{R}^3$ (ii) are linearly independent, so, yes, they form a basis for $\mathbb{R}^3$.

$\square$

Comment

The two previous examples give us two *different* bases for $\mathbb{R}^3$.
But:

$$\mathbb{R}^3$$

Basis #1: standard basis
$(1,0,0), (0,1,0), (0,0,1)$

Notice: 3 vectors.

Basis #2:
$(1,1,1), (1,2,3), (1,0,0)$

Notice: 3 vectors.

Thus we can say: "$\mathbb{R}^3$ has dimension 3."

Dimension

Video 12.1.3

Definition ('Dimension of a subspace').

Consider some Euclidean space $\mathbb{R}^n$ and some subspace $V \subset \mathbb{R}^n$.

The **dimension** of V is defined to be the number of vectors in any choice of basis B for V. In other words:

$$\dim V = (\#\, of\ elements\ in\ B).$$

Definition ('Dimension of a subspace').

Consider some Euclidean space $\mathbb{R}^n$ and some subspace $V \subset \mathbb{R}^n$.

The **dimension** of V is defined to be the number of vectors in any choice of basis B for V. In other words:

$$\dim V = (\# \, of \ elements \ in \ B).$$

There is an important issue here. We have defined dimension by first choosing a basis and then counting it. Whenever we define something by making a choice in mathematics we have to make sure that different choices don't give different answers.

*We need to know we **always get the same answer** no matter which basis we choose.*

The next theorem guarantees that.

Theorem ('Number of elements in a basis').

Consider some Euclidean space $\mathbb{R}^n$ and some subspace $V \subset \mathbb{R}^n$.

Let B_1 and B_2 be two different bases for V.

Then:

$$(\# \, of \ vectors \ in \ B_1) = (\# \, of \ vectors \ in \ B_2).$$

This theorem explains why dimension is a well-defined number associated to a subspace.

Comments on proof:

We won't discuss the proof of this important theorem in this course but you should prove it in a more general setting in Linear Algebra II.

The proof is elementary but lengthy: you show you can always turn the first given basis B_1 into the second given basis B_2 by a sequence of replacements of single vectors on the list:

$$B_1 \to \underbrace{\beta_1 \to \cdots \to \beta_k}_{\text{A sequence of intermediate bases all with the same number of elements.}} \to B_2.$$

Finding a basis for a subspace.

Video 12.1.4

So to calculate the dimension of a subspace (and for many other purposes) we will need to be able to compute a basis for a subspace V.

The way to do this depends on how the subspace is described to us. There are two main situations we need to learn to deal with.

Situation #1: When the subspace V is described to us as the span of some set of vectors S.

Situation #2: When the subspace V is described to us as the set of solutions to some system of homogeneous linear equations.

Situation #1: ***reducing a spanning set.***

Just say a subspace $V \subset \mathbb{R}^n$ is described to us by giving a list of spanning vectors S:

$$\vec{v}_1, \dots, \vec{v}_k.$$

In other words: V is the set of all possible linear combinations of the vectors on the list S. In other words: $V = span(S)$.

The problem: How to compute a basis for V?

We'll meet two versions of the solution.

- The first is nice because it is intuitive and easy to understand why it works.
- The second version will be an efficient matrix algorithm which does the same thing.

The problem: How to compute a basis for

$$V = span\{\vec{v}_1, \dots, \vec{v}_k\}?$$

First version of the solution:

1. Look for a vector $\vec{v}_i$ on this list which is a linear combination of the other vectors on this list:
$$\vec{v}_i = c_1\vec{v}_1 + \cdots + c_{i-1}\vec{v}_{i-1} + c_{i+1}\vec{v}_{i+1} + c_k\vec{v}_k$$
2. When you remove this vector (denoted $\widehat{\vec{v}_i}$) you get a smaller list which also spans V:
removing it from the list doesn't change the subspace it builds
$$span\{\vec{v}_1, \dots, \widehat{\vec{v}_i}, \dots, \vec{v}_k\} = span\{\vec{v}_1, \dots, \vec{v}_i, \dots, \vec{v}_k\} = V$$
3. Go back to step 1 using the shorter list.
4. Keep removing vectors like this until there are no such vectors left (in other words, until the set is linearly independent). The list you end up with is a basis for V.

Example:

Consider a subspace $V \subset \mathbb{R}^4$ described as the span of the list

$$(1,3,-2,0), (1,1,0,1), (1,5,-4,-1).$$

What is a basis for this subspace?

Solution:

We are going to look and see if any of the vectors on this list can be expressed as linear combinations of the other vectors on the list. If there are such vectors we remove them and try again.

We can determine such vectors by first solving for the possible c_1, c_2, c_3 such that

$$c_1(1,3,-2,0) + c_2(1,1,0,1) + c_3(1,5,-4,-1) = (0,0,0,0).$$

→ if there are solutions besides the trivial solution, then one of the vectors is a combination of other vectors

Solution (continued):

We are solving for the c_1, c_2, c_3 such that

$$c_1(1,3,-2,0) + c_2(1,1,0,1) + c_3(1,5,-4,-1) = (0,0,0,0).$$

This equation is equivalent to the following system of linear equations:

$$\begin{cases} c_1 + c_2 + c_3 & = & 0 \\ 3c_1 + c_2 + 5c_3 & = & 0 \\ -2c_1 + 0c_2 - 4c_3 & = & 0 \\ 0c_1 + c_2 - c_3 & = & 0 \end{cases}.$$

Solve:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 3 & 1 & 5 & 0 \\ -2 & 0 & -4 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

This gives solution: $c_3 = s, c_2 = s, c_1 = -2s$.

Solution (continued):

We just calculated that for every $s \in \mathbb{R}$ the following equation is true:

$$(-2s)(1,3,-2,0) + s(1,1,0,1) + s(1,5,-4,-1) = (0,0,0,0).$$

In particular if we set $s = 1$ we deduce:

$$(-2)(1,3,-2,0) + 1(1,1,0,1) + 1(1,5,-4,-1) = (0,0,0,0).$$

We can rearrange this to express the third vector in terms of the first two:

$$(1,5,-4,-1) = 2(1,3,-2,0) - (1,1,0,1).$$

Solution (continued):

So we can now cross the third vector off the list:

$$(1,3,-2,0), (1,1,0,1), \cancel{(1,5,-4,-1)}$$

Because $(1,5,-4,-1) = 2(1,3,-2,0) - (1,1,0,1)$ we can cross $(1,5,-4,1)$ off the list and end up with a smaller list giving the same span. (In other words: the set of all possible linear combinations will be exactly the same.)

$$(1,3,-2,0), (1,1,0,1)$$

builds the same subspace!

Solution (continued):

So then we go back to step one and ask: Is it possible to express one of the vectors on the following list in terms of the other vectors?

$$(1,3,-2,0),(1,1,0,1)$$

In this case there are only two vectors. In this case it is obvious to see that it is impossible to express one of the two vectors as a linear combination of the other one. (They would have to be proportional.)

Thus:

- This list is linearly independent.
- The span of this list is V.

Thus: the list $(1,3,-2,0),(1,1,0,1)$ is a basis for V. $\square$

Example:

Video 12.1.5

Consider a subspace $V \subset \mathbb{R}^3$ described as the span of the list

$$(1,1,0),(1,0,2),(2,1,2),(2,2,0).$$

What is a basis for this subspace?

Solution:

We start by finding all possible solutions to the vector equation $(\star)$:

$$c_1(1,1,0)+c_2(1,0,2)+c_3(2,1,2)+c_4(2,2,0)=(0,0,0).$$

The system is:

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 2 & 0 \\ 0 & 2 & 2 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 2 & 2 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Solution (continued):

We deduce that the set of solutions to the equation $(\star)$

$$c_1(1,1,0)+c_2(1,0,2)+c_3(2,1,2)+c_4(2,2,0)=(0,0,0)$$

is given by:

$$\begin{cases} c_1 &= -r-2s \\ c_2 &= -r \\ c_3 &= r \\ c_4 &= s \end{cases}.$$

Now we can try and reduce this list by asking: are any of the vectors on this list linear combinations of the other vectors on this list?

Solution (continued):

$$(1,1,0),(1,0,2),(2,1,2),\cancel{(2,2,0)}$$

If we set $r=0$ and $s=1$ into $(\star)$ we deduce

$$(2,2,0)=2(1,1,0)$$

so we can erase $(2,2,0)$ from the list.

$$(1,1,0),(1,0,2),\cancel{(2,1,2)}$$

If we set $r=1$ and $s=0$ into $(\star)$ we deduce

$$(2,1,2)=(1,1,0)+(1,0,2).$$

so we can erase $(2,1,2)$ from the list.

$(1,1,0),(1,0,2)$ → basis for span(S)

This pair of vectors is obviously linearly independent.
Thus: $(1,1,0),(1,0,2)$ is a basis for V. $\square$

Formulating this procedure as a general matrix algorithm.

Video 12.1.6

The computational problem:

Given a list of vectors $\vec{v}_1, \dots, \vec{v}_k$ in $\mathbb{R}^n$ reduce this list to get a basis for $V = span\{\vec{v}_1, \dots, \vec{v}_k\}$.

The input data:

- A positive integer n and a positive integer k.
- A list of k vectors in $\mathbb{R}^n$:

$$\begin{array}{ccc} \vec{v}_1 & = & (v_{11}, \dots, v_{1n}) \\ & \vdots & \\ \vec{v}_k & = & (v_{k1}, \dots, v_{kn}) \end{array}$$

The algorithm:

Step 1

Construct a matrix by making the vectors $\vec{v}_1 \dots \vec{v}_k$ the columns of the matrix.

$$\begin{bmatrix} \uparrow & & \uparrow \\ \vec{v}_1 & \cdots & \vec{v}_k \\ \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} v_{11} & \cdots & v_{k1} \\ \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} \end{bmatrix}.$$

Step 2

Do Gaussian elimination to put this matrix into row echelon form.

The algorithm (continued):

Step 3

Look for the columns in this matrix with leading entries. The corresponding vectors ***in the original list*** of vectors form a basis for $V = span\{\vec{v}_1, \dots, \vec{v}_k\}$.

if no leading entry in that column, that vector in that column is not in the basis

Problem:

Explain why this algorithm works.

Video 12.1.8

Situation #2: ***subspace given as a solution set of a homogeneous system.***

Example:

Consider the following system of linear equations:

$$\begin{cases} x_1 + x_2 - x_3 - x_4 & = & 0 \\ 3x_1 - x_2 + 2x_3 - x_4 & = & 0. \\ 4x_1 + 0x_2 + x_3 - 2x_4 & = & 0 \end{cases}$$

This is a homogeneous system of equations. Thus the set of solutions $S \subset \mathbb{R}^4$ to the system is a subspace of $\mathbb{R}^4$.

What is the dimension of S?

Solution:

According to the definition of dimension, to find the dimension of S we need to:

1. Find a basis B for S.
2. Count the number of vectors in the basis B.

So how do we find a basis B? Well the first property of a basis is that it should form a spanning set for S. In other words: $S = span(B)$.

So let's see if we can find any spanning set for S. To do that we need to first understand S.

A way to do this is to solve for a parameterization of the solution set S.

Solution (continued):

So we'll start by solving for a parameterization of the set of solutions S.

The augmented matrix corresponding to the linear system is:

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 3 & -1 & 2 & -1 & 0 \\ 4 & 0 & 1 & -2 & 0 \end{array}\right].$$

We'll apply Gauss-Jordan elimination to solve this system:

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 0 & -4 & 5 & 2 & 0 \\ 0 & -4 & 5 & 2 & 0 \end{array}\right] \rightarrow\rightarrow \left[\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 0 & -4 & 5 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Solution (continued):

Continuing with Gauss-Jordan elimination:

$$\to \left[\begin{array}{cccc|c} 1 & 1 & -1 & -1 & 0 \\ 0 & 1 & -5/4 & -2/4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 1/4 & -1/2 & 0 \\ 0 & 1 & -5/4 & -1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

This matrix is in reduced row echelon form, so as usual, we can do back substitution to write down the set of solutions.

Following the usual procedure: there are two columns without leading entries. The corresponding variables x_3 and x_4 will become free parameters. Then we solve for the remaining variables x_1 and x_2 in terms of these free parameters.

Solution (continued):

We deduce that the set of solutions S is given by the following parameterization:

$$\begin{cases} x_1 &= -\frac{1}{4}s + \frac{1}{2}t \\ x_2 &= \frac{5}{4}s + \frac{1}{2}t \\ x_3 &= s \\ x_4 &= t \end{cases} \quad , s, t \in \mathbb{R}.$$

To see the spanning set clearly we should write down this parameterization in vector notation:

$$S = \left\{ \left(-\frac{1}{4}s + \frac{1}{2}t, \frac{5}{4}s + \frac{1}{2}t, s, t \right), s, t \in \mathbb{R} \right\}.$$

<u>Solution (continued):</u>

This can be written more clearly by separating the variables:

$$S = \left\{ s\left(-\frac{1}{4}, \frac{5}{4}, 1, 0\right) + t\left(\frac{1}{2}, \frac{1}{2}, 0, 1\right), s, t \in \mathbb{R} \right\}.$$

What is a basis B for S ?

We guess that B is given by the list consisting of these two vectors: $\left(-\frac{1}{4}, \frac{5}{4}, 1, 0\right), \left(\frac{1}{2}, \frac{1}{2}, 0, 1\right)$.

As usual there are two things to check:

1. Does this form a spanning set for S?
2. Is this list linearly independent?

<u>Solution (continued):</u>

Question #1: Why does this list span S?

This is exactly the reason we chose this list B. Because

$$S = \left\{ s\left(-\frac{1}{4}, \frac{5}{4}, 1, 0\right) + t\left(\frac{1}{2}, \frac{1}{2}, 0, 1\right), s, t \in \mathbb{R} \right\}$$

it is obvious that every vector in S is a linear combination of these two vectors: $\left(-\frac{1}{4}, \frac{5}{4}, 1, 0\right), \left(\frac{1}{2}, \frac{1}{2}, 0, 1\right)$.

Solution (continued):

Question #2: Why is this list linearly independent?

We can just check this directly in the way we have learnt. We need to show that the only numbers c_1, c_2 that solve the following equation

$$c_1\left(-\frac{1}{4}, \frac{5}{4}, 1, 0\right) + c_2\left(\frac{1}{2}, \frac{1}{2}, 0, 1\right) = (0,0,0,0)$$

is the trivial solution $c_1 = c_2 = 0$.

This is easy to see: If we look at the third position in this vector equation get the scalar equation $c_1 + 0c_2 = 0$. Thus we deduce $c_1 = 0$.

Also if look at the fourth position in this vector equation we see the equation $0c_1 + c_2 = 0$. Thus $c_2 = c_1 = 0$.

Thus this list is linearly independent.

Solution (Conclusion.):

We investigated the subspace of $\mathbb{R}^4$ described as the set of solutions $S \subset \mathbb{R}^4$ to the system of linear equations

$$\begin{cases} x_1 + x_2 - x_3 - x_4 & = & 0 \\ 3x_1 - x_2 + 2x_3 - x_4 & = & 0. \\ 4x_1 + 0x_2 + x_3 - 2x_4 & = & 0 \end{cases}$$

By solving for a parameterization of S we guessed that a basis B for S is given by

$$B = \left(-\frac{1}{4}, \frac{5}{4}, 1, 0\right), \left(\frac{1}{2}, \frac{1}{2}, 0, 1\right).$$

Then we proved that this B is indeed a basis for S by showing (i) that every vector in S is a linear combination of the vectors in B and (ii) that B is linearly independent.

Thus: $\dim S = (\#\text{ of vectors in } B) = 2.$ $\square$

Summary of this method:

1. Start with a homogeneous system of linear equations. Ask: what is a basis for S, the set of solutions of the system?
2. Solve for a parameterization of S in vector notation.
3. Separate the formula you get into a sum of separate vectors, one for each for each of the variables.
4. It turns out the list of vectors you get from this is exactly a basis for S.
5. To get the dimension of S just count the vectors on this list.

Exercise:

Video 12.1.9

Consider the following homogeneous linear system in 5 variables:

$$\begin{cases} x_1 + x_2 + x_3 + x_4 + x_5 & = & 0 \\ x_1 + x_2 + 2x_3 + 4x_4 + 6x_5 & = & 0 \end{cases}.$$

Because the system is homogeneous we know that the set of solutions S is a subspace of $\mathbb{R}^5$.

Determine a basis for S and then state $\dim S$.

Solution:

First we need to find a parameterization of the set of solutions. We can employ Gauss-Jordan elimination to obtain that.

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 2 & 4 & 6 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 3 & 5 & 0 \end{array}\right]$$

And then:

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & -2 & -4 & 0 \\ 0 & 0 & 1 & 3 & 5 & 0 \end{array}\right].$$

Then back-substitution leads to the general solution:

$$\{(-r+2s+4t, r, -3s-5t, s, t), r, s, t \in \mathbb{R}\}.$$

Solution (continued):

Then back-substitution leads to the general solution:

$$\{(-r+2s+4t, r, -3s-5t, s, t), r, s, t \in \mathbb{R}\}.$$

When we separate parameters this expression becomes:

$$\begin{gathered} r(-1,1,0,0,0) \\ +s(2,0,-3,1,0) \\ +t(4,0,-5,0,1). \end{gathered}$$

Thus a basis for the solution space S is:

$$(-1,1,0,0,0), (2,0,-3,1,0), (4,0,-5,0,1).$$

There are three vectors in this basis, so: $\dim S = 3$.

$\square$